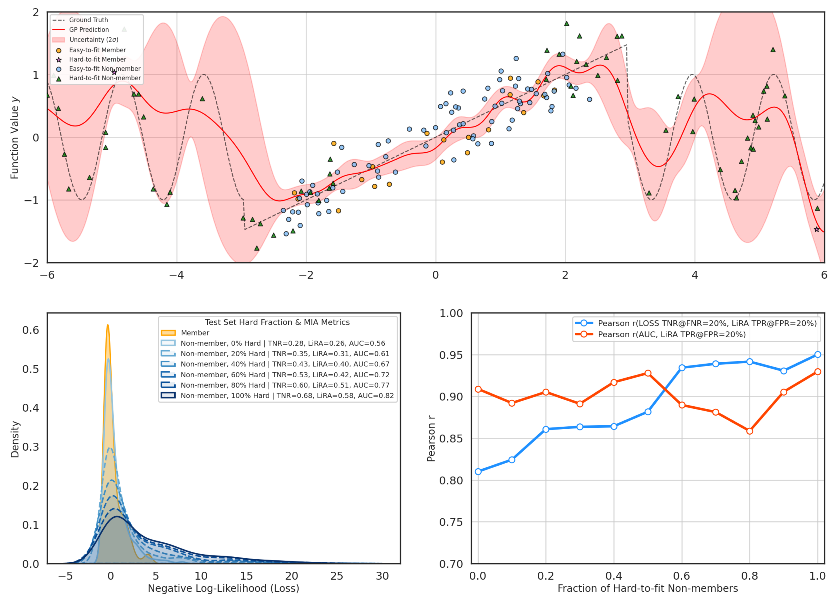

Figure 1: Relationship between loss-distribution shape and the best loss-based proxy for model-level MIA risk. Top: toy GP regression setting with easy-to-fit and hard-to-fit members (orange) and non-members (blue), showing how sparse, high-uncertainty regions induce large losses for hard non-members. Bottom-left: as the fraction of hard-to-fit non-members increases, the non-member loss distribution becomes increasingly heavy-tailed, increasing both LiRA TPR and the predictive power of tail-sensitive metrics. Bottom-right: Pearson correlation with LiRA TPR shows that LOSS AUC is more stable when the loss distributions are relatively symmetric, whereas LOSS TNR becomes the stronger predictor once the non-member distribution is sufficiently heavy-tailed.

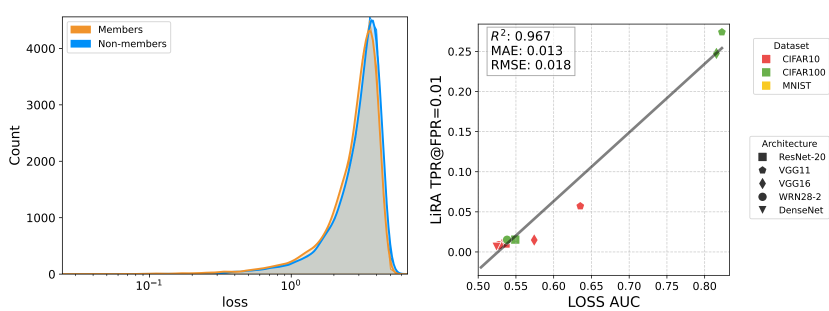

Figure 2: Left: Member and non-member distributions for ResNet-20 trained on CIFAR100 using RelaxLoss with $\alpha = 3$; Right: LOSS AUC predicts LiRA TPR@FPR=0.01 across varied model architectures and datasets. The linear relationship (solid grey line) exhibits strong performance with RMSE=0.018.